

The future looks dark: improving high contrast imaging with data-driven predictive wavefront control

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Wind-Driven Halo in Coronagraphic Images

When the correction speed of an AO (adaptive optics) system cannot keep up with atmospheric winds, coronagraphic images can be impacted by a wind-driven halo, where light is blown directionally along the wind layer into what would otherwise be a dark-zone prime for the direct imaging of exoplanets.

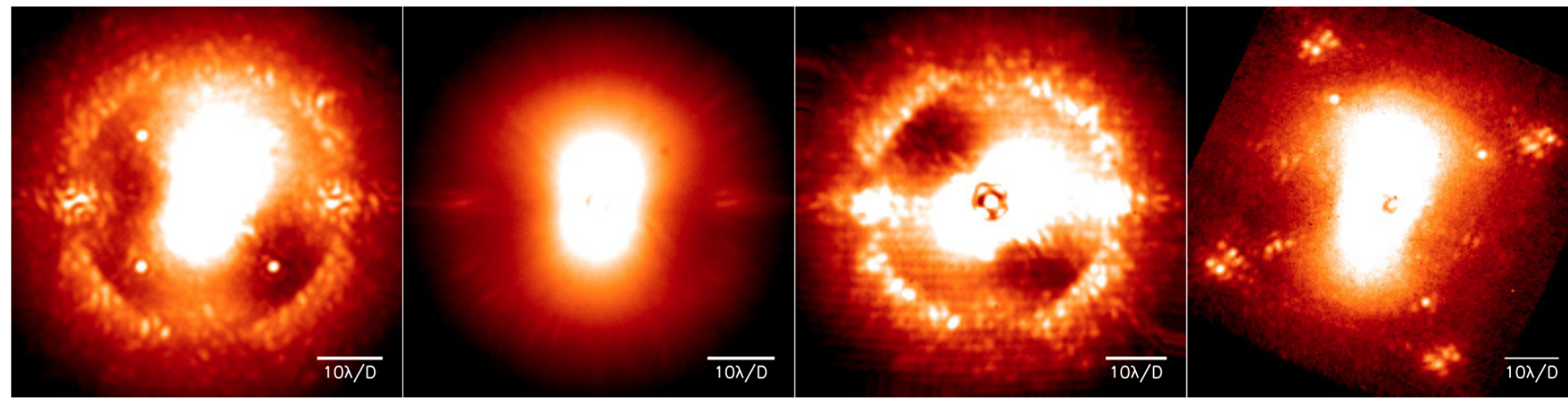


Figure 1: Figure from [1], showing the wind-driven halo in coronagraphic SPHERE and GPI images. Left: SPHERE-IRDIS, H2 band. Middle Left: SPHERE-IFS, YH mode. Middle Right: SPHERE-IRDIS, H band. Right: GPI, YH mode.

Empirical Orthogonal Functions

Empirical orthogonal functions (EOF [2]) builds a linear filter $\mathbf{F}$ by associating previous states of wavefront sensor information (or history vectors $\mathbf{h}$) into a training matrix $\mathbf{D}$, and comparing those to known future states $\mathbf{P}$. This filter matrix predicts future states in real time, having learned the wind speeds and other linear system errors.

$$\min ||\mathbf{D}^T \mathbf{F}^T - \mathbf{P}^T||^2 \quad (1)$$

$$\mathbf{F} = ((\mathbf{D}^T)^{\dagger} \mathbf{P}^T)^T \quad (2)$$

$$\mathbf{F} = \mathbf{P} \mathbf{D}^T (\mathbf{D} \mathbf{D}^T + \alpha \mathbf{I})^{-1} \quad (3)$$

$$\text{prediction} = \mathbf{F} \mathbf{h} \quad (4)$$

(where α is a regularization constant used for a least-squares pseudo-inverse.)

At every iteration of the real-time-controller, we multiply the filter matrix $\mathbf{F}$, by the history vector $\mathbf{h}$ (represented visually in Figure 2) to predict an iteration one lag-time in the future. This full logic is visualized in Figure 3.

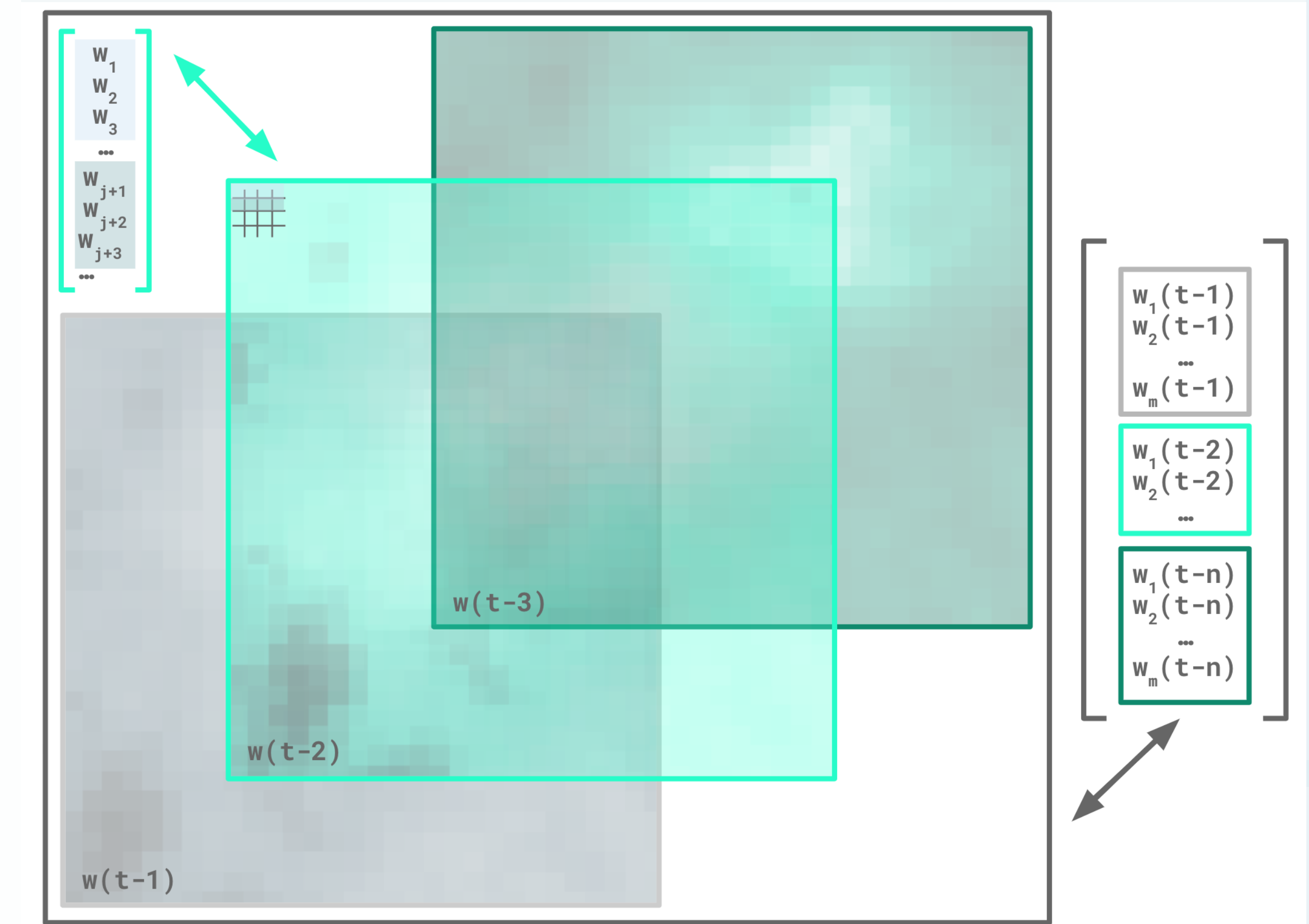


Figure 2: The history vector contains n frames of wavefront sensor data flattened into a single vector.

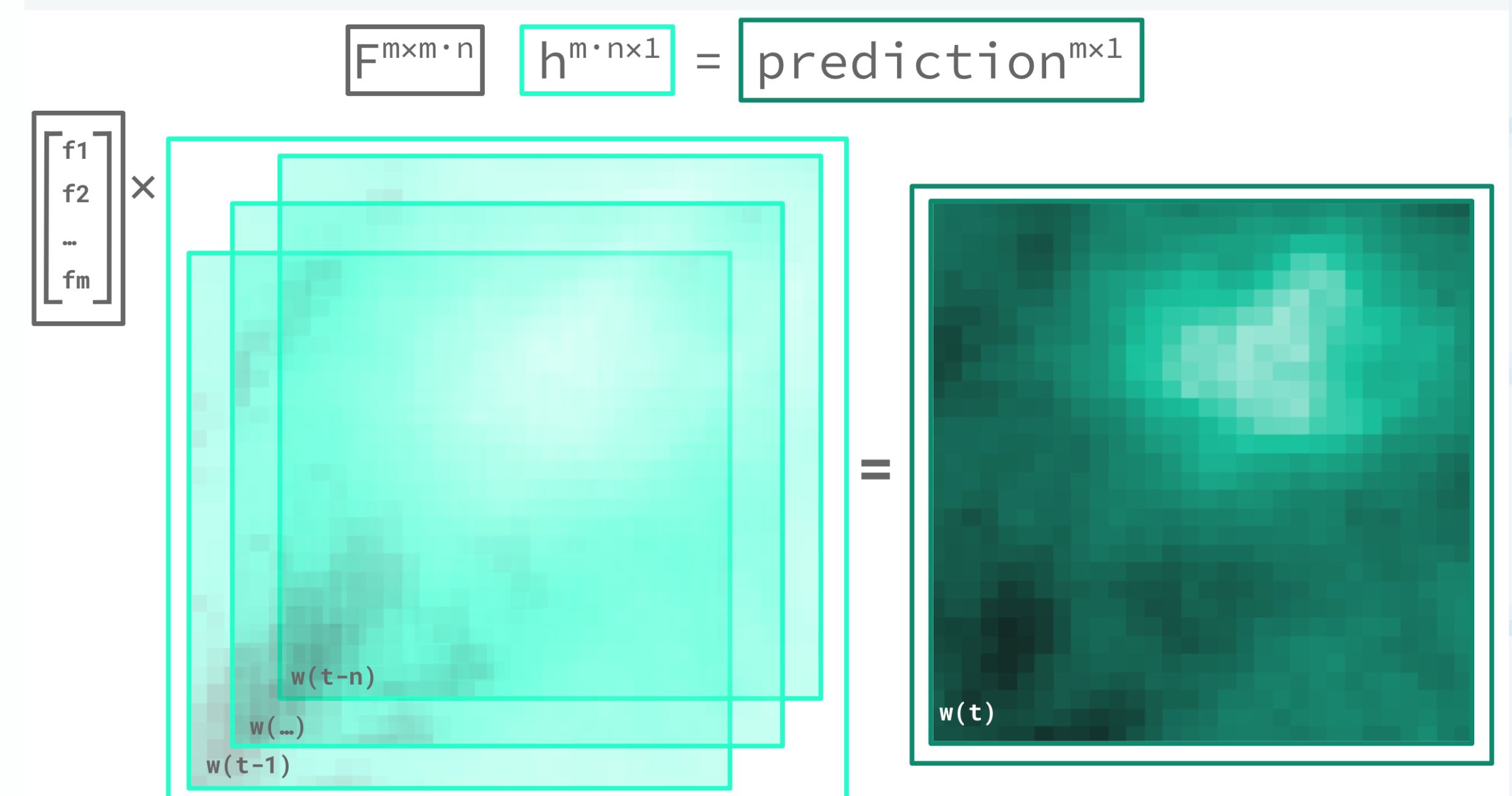


Figure 3: To predict a future iteration with m modes, we multiply a filter matrix that is m by $m \cdot n$ (with n points in time for each mode) by the current history vector $\mathbf{h}$.

While predictive control has been demonstrated on-sky both at Keck [3] and Subaru [4], little has been done in the way of filter parameter optimization.

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Abstract

The direct imaging and characterization of exoplanets requires extreme adaptive optics (XAO), achieving exquisite wavefront correction (upwards of 90% Strehl) over a narrow field of view (a few arcseconds.) For these XAO systems bandwidth error is often a leading term in the error budget, wherein the wavefront evolves faster than the lag between wavefront sensing and control – for atmospheres driven by fast wind layers, this can result in a wind-driven halo in the coronagraphic dark-zone, limiting sensitivity to faint, close-in companions. This lag-time is often limited by the wavefront sensor exposure time, especially in the case of lower magnitude guidestars. Predictive control mitigates the bandwidth error by predicting the shape of the wavefront by time the system correction is applied. One such method of prediction is empirical orthogonal functions (EOF), wherein previous states in the wavefront sensor history are used to learn linear correlations with a minimization problem. This method has been demonstrated on-sky at Subaru/SCEXAO and Keck/NIRC2, but has yet to be optimized. With this work as a starting point, we explore the optimal parameter space for implementing EOF on-sky, study its stability under varying atmospheric parameters, and discuss future paths for facilitization of predictive control. This work not only offers a pathway to optimize Keck and Subaru observing, but also acts as a pathfinder for predictive control methods with extremely large telescopes.

Preliminary EOF Demonstration on the SEAL Testbed

We implemented EOF on the SEAL (Santa Cruz Extreme Adaptive Optics Laboratory [5]) testbed, using a high speed Thorlabs Shack-Hartmann Wavefront Sensor with 640 slopes and a Boston Microachines MEMS deformable mirror with 24 actuators across the pupil. Initial results show improvement over a classic integrator both in reduction of wavefront error (Figure 4), and in the the focal plane PSF images (Figure 5).

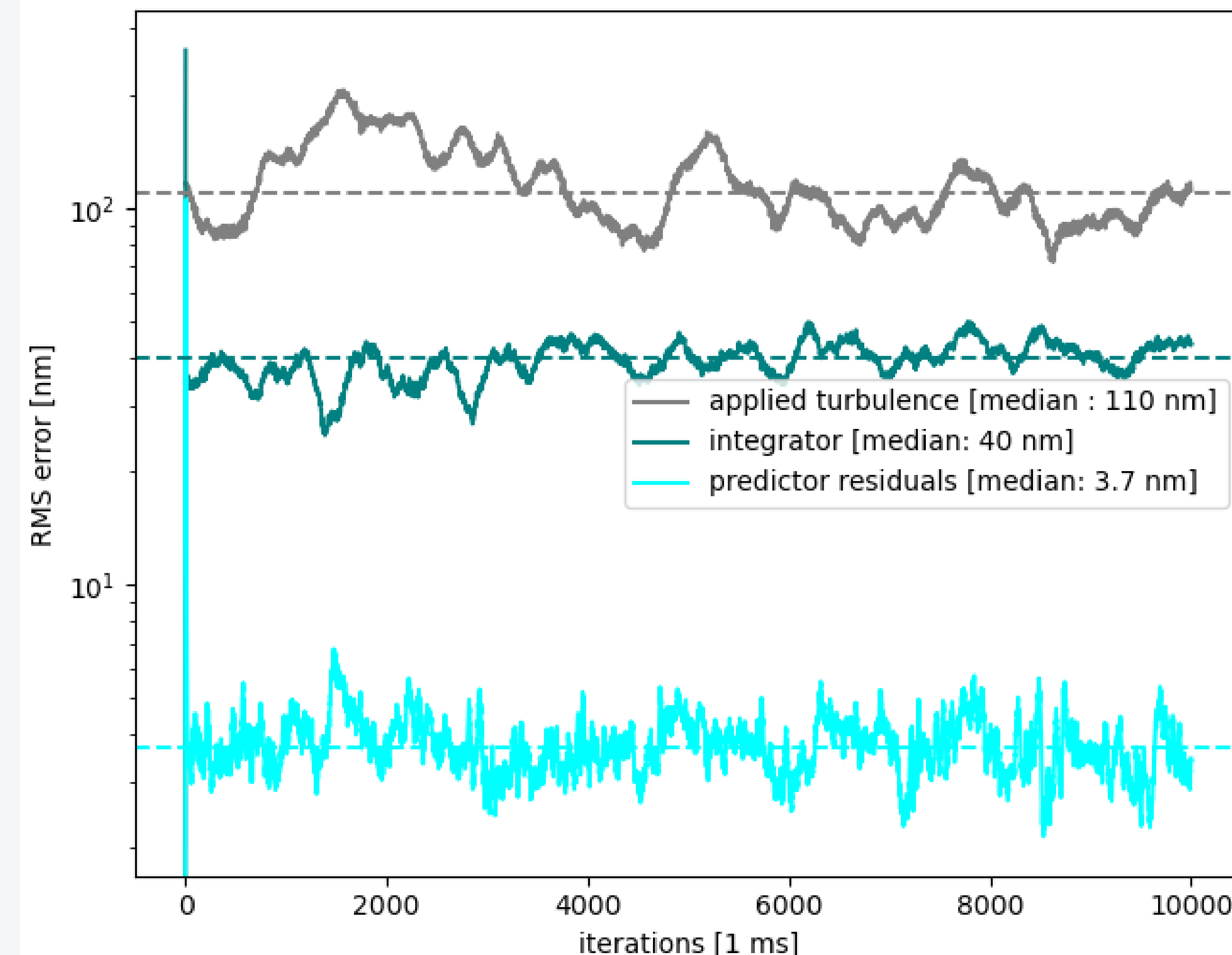


Figure 4: Bench results of an EOF controller as compared to a classic integrator. RMS wavefront is estimated from the SHWFS residuals as converted to DM space. We simulated a single layer atmosphere with an r_0 of 10cm and a 10 m/s wind layer for a 10 meter telescope. Turbulence was scaled down to the linear regime of the DM, and the applied and corrected with the DM for 10000 iterations in “fake real time” (i.e., real-time from the perspective of the data inputs and algorithm, but not yet optimized to run in real time.)

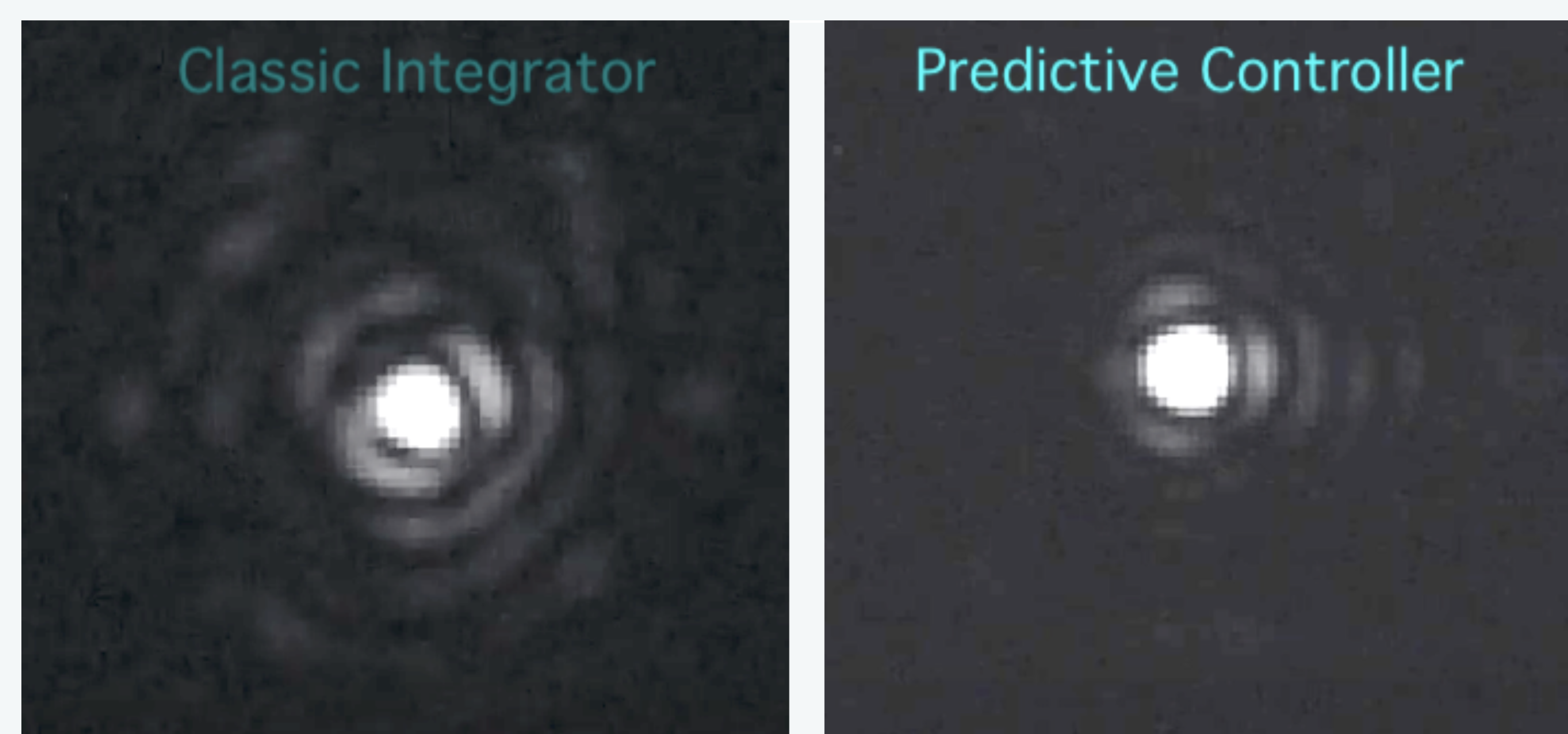


Figure 5: PSFs of predictive control and a classic integrator side-by-side. Note: these are preliminary results without robust image calibration and serve only as an initial visual comparison.

Optimizing Filter Parameters

In simulation, we find that the optimal parameter changes based on wind velocity and Fried parameter r_0 , implying that the current approach of using static sizes for the generation of these filters is suboptimal. Figure 6 shows performance estimates of a predictive controller for two different sets of atmospheric conditions based on varying filter parameters.

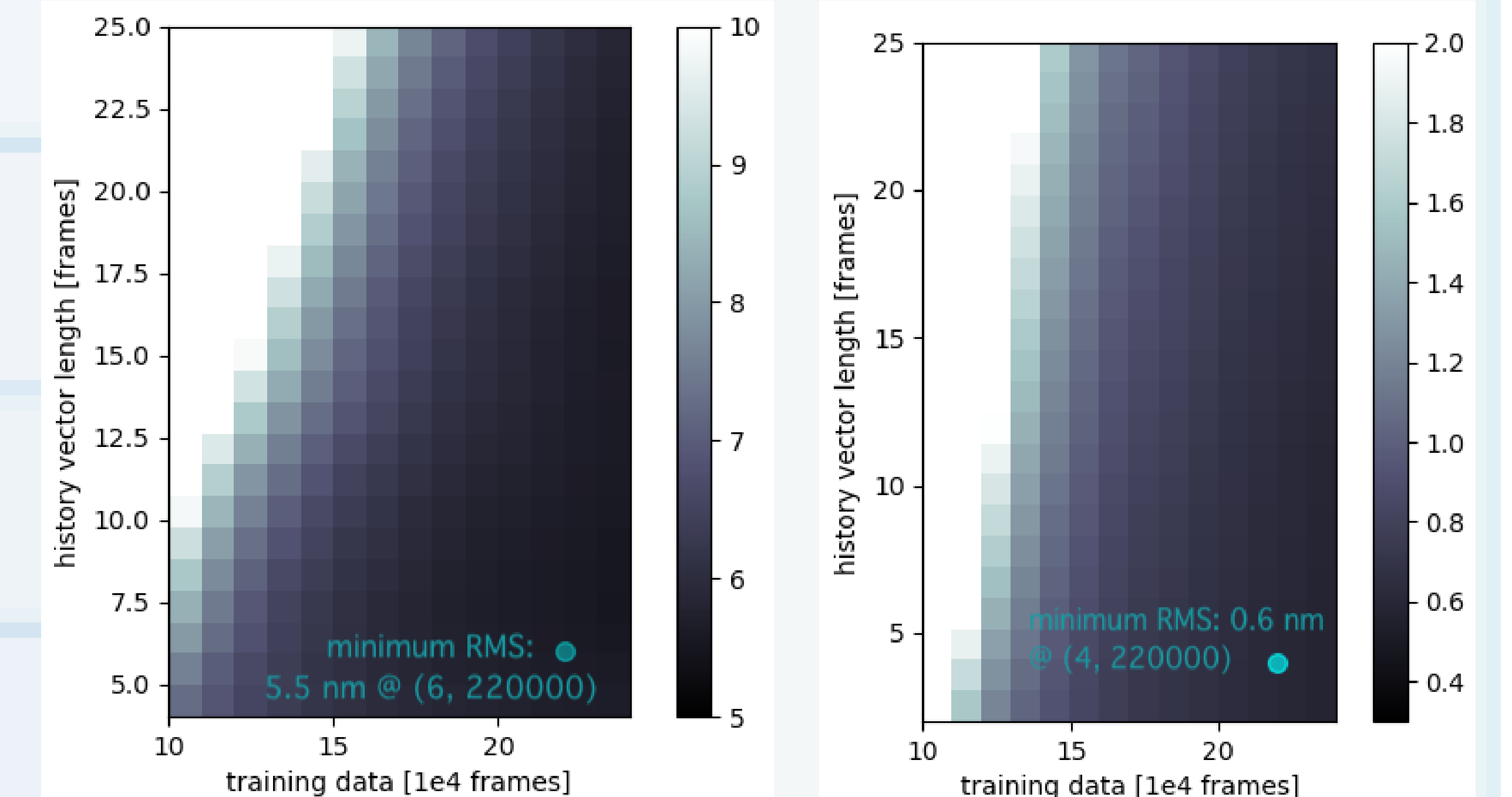


Figure 6: Left: Multi-layer atmosphere with wind velocities and a C_n^2 profile based on Mauna Kea conditions [6]. Right: Single layer atmosphere simulated at 7 m/s at an r_0 of 15 cm. Both simulations run with a loop speed of 2kHz. The colormap shows median RMS error in nm from each parameter combination. The blue point indicates the optimal solution. Differing atmospheric conditions change the optimal filter parameters.

On-the-fly Optimization with Simulated Annealing

Because building grids of this nature is computationally costly, we investigate a simulated annealing algorithm to optimally sample the parameter space. For a 22x23 grid simulated annealing could consistently find within 1 nm RMS of the optimal value within 10 iterations. Figure 7 shows an example of the steps simulated annealing tested and accepted for a representative run.

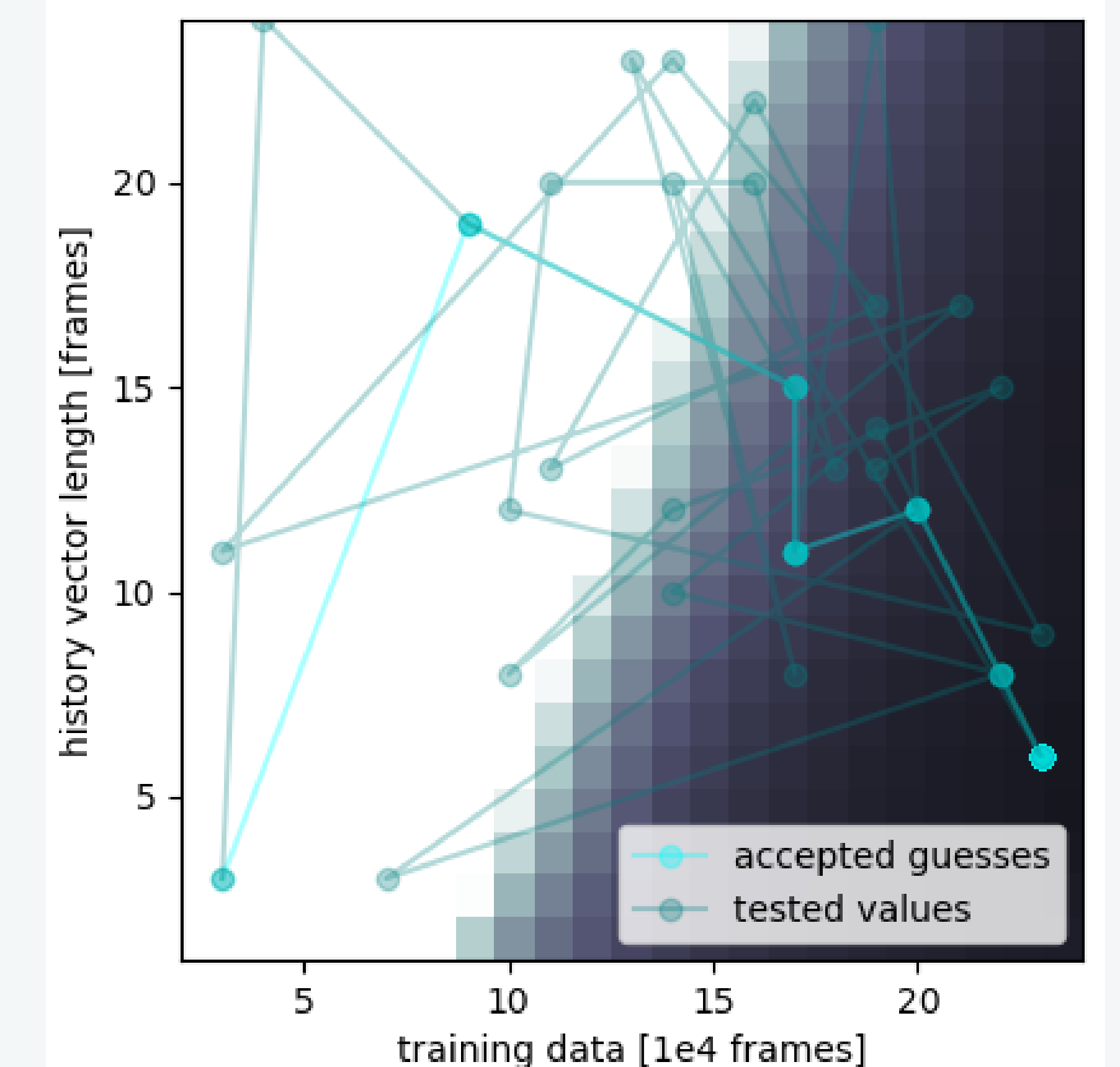


Figure 7: Example of the simulated annealing algorithm applied to the multi-layer turbulence profile above. The algorithm was able to find within 1 nm RMS of the optimal value with 3 iterations, and found the optimal value within 100 iterations, as compared to running the full grid with 506 values.

Next Steps

- Run a similar parameter grid on the SEAL testbed, including optimal α parameter.
- Look at the impact of these filter parameters in classic un-obscured and coronagraphic PSFs.
- Make predictions for optimal parameters and test on-sky with the Keck-II AO and/or SCEXAO.
- Test simulated annealing as a quasi-real time way to optimize filter parameters on-sky.

References

- 1 Cantalloube et al., 2018: *Origin of the asymmetry of the wind driven halo observed in high-contrast images*
- 2 Guyon & Males 2017: *Adaptive Optics Predictive Control with Empirical Orthogonal Functions (EOFs)*
- 3 van Kooten et al. 2022: *Predictive wavefront control on Keck II adaptive optics bench: on-sky coronagraphic results*
- 4 Guyon et al. 2020: *Adaptive optics real-time control with the compute and control for adaptive optics (Cacao) software framework*
- 5 Jensen-Clem et al. 2021: *The Santa Cruz Extreme AO Lab (SEAL): design and first light*
- 6 Neyman 2004: *Atmospheric Parameters for Mauna Kea*

